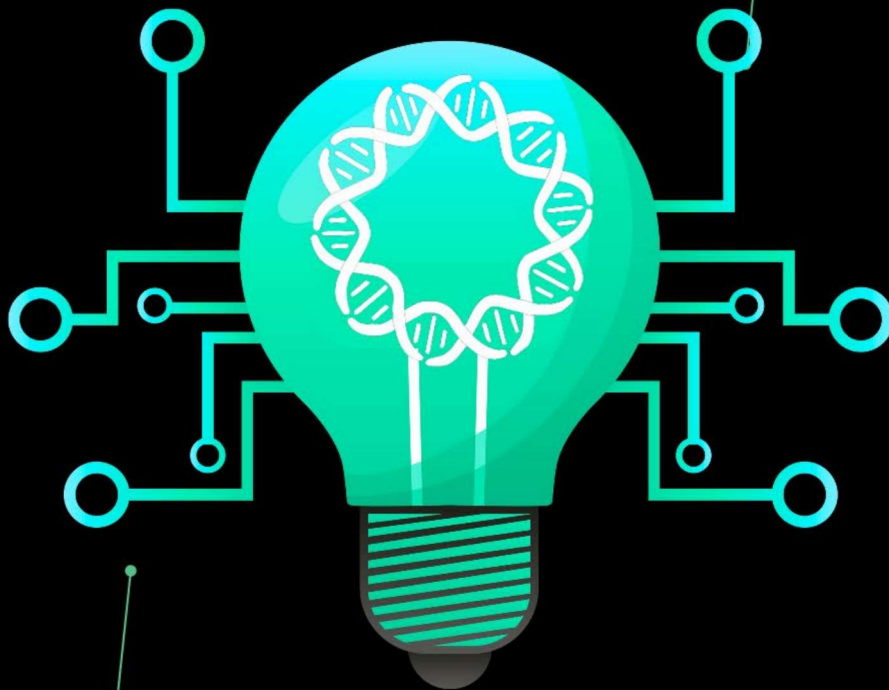




Harvard Undergraduate
OpenBio
Laboratory

P R E S E N T S



S R I

S T U D E N T R E S E A R C H I N S T I T U T E

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Table of Contents

ORIGINAL RESEARCH

CUPNavigator: A multi-model CDSS for tumor origin prediction, diagnostic planning, and therapy recommendations in cancer of unknown primary

Saicharan Vellanki and Salomon Kobongo

DROID: Dual-headed Representation for Omics-based Identification of SARS-CoV-2 Infection and Cell type via Deep Learning

Enan Khan and Tuan Pham

Integrative lipidomics and network pharmacology reveal the mechanisms of cannabidiol in the treatment of acute myeloid leukemia

Aiden Yoon and Tomiwa Oyedokun

Bi-stable Auxetic Stent Design for Customized Vascular Conformability

Tarun Malarvasan and Hrishikesh Kanade

Decoding the dark proteome: Deep learning-enabled discovery of druggable enzymes in *Wuchereria bancrofti*

Shawnak Shivakumar and Jefferson Hernandez

LITERATURE REVIEW

The potential of CRISPR-Cas9 gene editing technology to enhance CAR T-cell therapy as a targeted treatment for leukemia

Adwita Jagannathan and Daniella Lu

Reassessing the Undruggable: A Systematic Review and Meta-Analysis of KRAS G12C Inhibitor Efficacy in 2,031 Cancer Patients

Suchir Ryali and Andrew Evans

Interplay of Microglia and the Glymphatic system in Alzheimer's Disease A Systematic Review on the Crosstalk of Microglia and the Glymphatic System in Alzheimer's Disease

Dhriti Kantipudi and Hannah Yoder

The dialect of chromatin in chronic pain and its distortion under opioid exposure

Sahana Sivam and Carlos Rivera

The role of stress-related biomolecules in pancreatic ductal adenocarcinoma: A tidal wave of consequences

Vivian Zhao and Brielle Patlin

CONFERENCE PROCEEDINGS

Letter from the Harvard Undergraduate OpenBio Laboratory Student Research Institute Team

Dear Readers,

It is a joy and an honor to share this issue of STEM Fellowship Journal, encapsulating the inquisitiveness, dedication, and commitment to science of the Student Research Institute (SRI) Summer 2025 cohort. Within these pages, you will find the work of students who remind us why science matters and why access to it must be protected and expanded. The student researchers of this second cohort of SRI have shown us that curiosity knows no boundaries of geography, background, or circumstance.

SRI was created with the simple belief that every young person with a question burning inside them deserves the chance to explore it. Every student who dreams of research deserves a seat at the table, regardless of whether their school has a lab, their family can pay for a program, or they even know what “research” looks like. This summer, more than 2,000 students from 38 countries reached out with that dream. One hundred and fifteen were selected to embark on an eight-week journey guided by mentors from 32 universities. Together, they explored fields from CRISPR-CAR T-cell therapy to tumor origin prediction. They read, they asked, they struggled, they discovered. And in the end, they produced papers and posters that reflect not only what they learned but who they are becoming as scientists.

None of this would have been possible without the extraordinary mentors who gave their time and care. Balancing their own demanding research, they chose to teach, to guide, and to believe in the potential of students they had never met before. Their generosity created spaces where curiosity could take root and where confidence could begin to grow. We are deeply grateful for their example.

The pages that follow are filled with the beginnings of scientific journeys. Yet they are more than first steps. They are reminders of what science can be when access is widened and voices once overlooked are invited in. They are proof that talent is everywhere, waiting for an open door.

As you read, we invite you not only to celebrate the accomplishments of these young researchers but also to imagine what it means for the future of science when they are given the tools to thrive. The discoveries they make in the years ahead may change fields we cannot yet name. What matters now is that they have begun, and that they know they belong here.

On behalf of the Harvard Undergraduate OpenBio Laboratory, we extend our heartfelt congratulations to this year’s students, our gratitude to their mentors, and our hope for all the journeys that lie ahead. The future of science is undoubtedly brighter because of them.

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4TH PLACE, ORIGINAL RESEARCH TRACK

Bi-stable Auxetic Stent Design for Customized Vascular Conformability

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ABSTRACT: Cardiovascular diseases remain the leading cause of mortality worldwide, often requiring vascular stents to restore blood flow. However, complications such as stent-adjacent stenosis—the re-narrowing of a blood vessel after stent placement—occur in 30% of stent patients. Restenosis often arises from the uniform radial expansion of conventional stents, which fails to account for irregular vascular geometries of stenotic regions and healthy regions, leading to uneven force distribution and damage to the vessel wall. This research proposes a stent design leveraging bi-stable auxetic mechanisms as well as patient-specific customizations to enhance performance and minimize complications. Auxetic lattice structures are characterized by a lateral extension when stretched, enabling the stent to conform to complex vascular geometries more naturally than conventional metal mesh stents by distributing radial forces evenly to reduce localized stress and vascular trauma. The bi-stable mechanism is used to ensure stability in both contracted and expanded configurations so that a vessel cannot re-narrow after placement. The stent's expanded configuration is customized through machine learning and patient-specific imaging data from MRI and CT angiography. A 3D reconstruction of the blood vessel is converted into a CAD model, allowing the auxetic lattice to be tailored to deliver variable radial forces to stenotic and healthy regions. Finite element analysis (FEA) was also used to optimize stress distribution and evaluate mechanical durability under pulsatile flow conditions. Preliminary results with computational models of the stent design demonstrate a reduction in restenosis, minimized over-compression, and enhanced vascular wall support.

KEYWORDS: stents, auxetics, patient-specific, machine-learning, FEA

INTRODUCTION

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, with an estimated 17.9 million deaths [1]. Many forms of CVD involve the blockage of arteries due to plaque buildup, which impairs blood flow and can lead to strokes [2]. A common treatment for arterial obstructions is a stent: a cylindrical mesh device that expands the pressing arterial plaque against the vessel wall to restore normal circulation [3]. Stents are commonly deployed using a balloon catheter in a minimally invasive procedure called percutaneous coronary intervention (PCI) [4]. Once positioned, the balloon is inflated to expand the stent.

Although the development of drug-eluting stents (DES) has reduced the rate of restenosis — the re-narrowing of the vessel after stent placement — this complication still occurs in 30-40% of patients [5,6]. Restenosis is often driven

by neointimal hyperplasia (excessive tissue growth inside the stent) and vessel remodeling. These can both be triggered by localized mechanical injury from stents [7,8]. This is because conventional stents expand uniformly along their length and are unable to conform to the branching angles and irregular shapes of blood vessels. This mismatch in geometry can lead to uneven radial force distribution which can produce localized high-stress regions on the vessel wall that promote inflammatory responses and vessel wall damage [9].

This work aims to address the limitations of conventional stents by combining bi-stable auxetics with patient-specific customization. This approach represents a shift from one-size-fits-all stent designs to precision-engineered vascular implants tailored to individual patient anatomy.

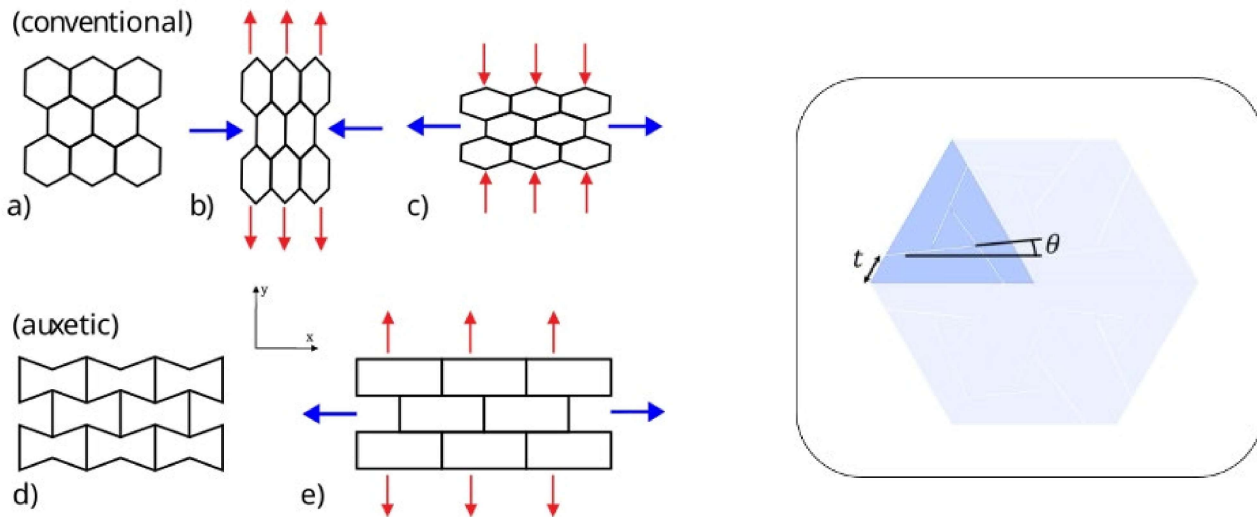


Figure 1. Left: Diagram of Conventional vs. Auxetic structures. Showcases the unique expansion properties of auxetics. Right: Illustrates an individual auxetic unit. Variable t is pictured on the left demonstrating a strut width. Variable θ is pictured in the middle demonstrating the angle between two struts.

METHODS

Auxetic Materials

Auxetic lattices exhibit a negative Poisson's ratio (NPR): when stretched longitudinally, they expand laterally rather than contracting [10,11].

In the context of a vascular stent, this geometric response means that radial expansion of the stent produces axial expansion of the lattice. This longitudinal-to-radial coupling is advantageous for addressing restenosis and stent-adjacent stenosis. Conventional stents rely on uniform radial expansion, which can over-compress healthy segments or insufficiently open severely stenotic regions, producing uneven contact pressure and local vessel trauma. In contrast, an auxetic stent can deliver a variable radial expansion profile along its length: regions corresponding to a stenosis undergo larger localized opening while adjacent healthy regions receive lower compressive force [12]. This improves contact conformity and reduces peak contact stresses on the arterial wall, thereby lowering the mechanical triggers of restenosis (e.g., endothelial damage and inflammation).

Bi-Stable Nature of Auxetics

Bi-stability refers to the property of having two different states. Bi-stable behavior is achieved by incorporating compliant hinge regions [13,14] within the auxetic lattice,

allowing the stent to have two mechanically stable states:

1. Contracted State: For catheter delivery, where the stent maintains a small diameter without requiring continuous constraint.
2. Expanded State: Post-deployment, where the stent remains fully open, resisting any arterial force.

The expansion rate of an auxetic material can be altered [15] using two variables within each auxetic unit: t and θ (Figure 1). A low t and θ value results in a low expansion ratio whereas a high t and θ value corresponds to a high expansion ratio.

To create a patient-specific stent, the stenotic region of the artery needs to be identified. For this purpose, a deep learning-based semantic segmentation model was developed using the MONAI (Medical Open Network for AI) framework.

A 2D U-Net Convolutional Neural Network (CNN) [16,17] architecture was used, leveraging MONAI's implementation optimized for biomedical image segmentation. The U-Net's encoder-decoder structure allows for effective feature extraction and precise pixel-wise classification, allowing the model to segment healthy, stenotic (area in which a stent needs to be placed), and thrombotic (areas of plaque buildup along an arterial wall) regions within the coronary arteries accurately (Figure 2).

The model was trained using the ARCADE dataset, consisting of 3,000 annotated X-ray Coronary Angiography

(XCA) images. This dataset was chosen due to the varying degrees of stent-adjacent stenosis across the images. The model was trained using the Adam optimizer with an initial learning rate of 0.001 as well as the Dice Loss function to address class imbalance.

Data augmentation methods like random rotations, horizontal and vertical flips, and intensity scaling were also applied during training to enhance model generalization. The ARCADE dataset was split into 80% training, 10% validation, and 10% testing subsets. Training continued until convergence with prevention efforts of overfitting by monitoring model performance monitored on the validation set.

Three-Dimensional Arterial Volume Approximation

Three-Dimensional Arterial Volume Approximation

Once the two-dimensional stenotic region has been segmented, a three-dimensional volume must be obtained to create a respective auxetic stent. To generate 3D models of coronary arteries for patient-specific stent design, a 3D U-Net segmentation model [18,19] was trained on the COCA (Coronary Calcium and Chest CT) dataset. This dataset includes annotated 3D CT scans that capture detailed coronary anatomy, including both healthy and stenotic vessel regions.

The COCA dataset consists of high-resolution chest CT volumes along with calcium buildup to simulate a typical stent-viable patient. Before training, scans underwent normalization and voxel spacing standardization to ensure consistent image quality. Noise reduction techniques were also applied to improve segmentation accuracy.

The network was trained using a combination of Dice Loss and Cross-Entropy Loss to handle class imbalance and optimize segmentation performance. Rotations and elastic deformations were also used for 3D data augmentation to enhance robustness to anatomical variability.

The model incorporates coordinate data derived from the previous 2D coronary artery segmentation model. The 2D stenotic region coordinates are spatially mapped and correlated with corresponding slices in the 3D CT volume, allowing the 3D reconstruction to accurately highlight stenotic vessel segments in 3D space.

After volumetric segmentation, the arterial cross-sections of the volumetric CT are aggregated into a 3D mask. Then,

the 3D mask is processed using the Kitware Visualization Toolkit (VTK) [20] to generate a triangulated surface mesh that represents the artery's volume. Surface smoothing is applied through the VTK to remove any remaining artifacts. The final 3D model consists of both healthy and stenotic vessel regions, informed by the integrated 2D stenosis data. Finally, the reconstructed artery mesh is exported into a CAD-compatible format (STEP).

Patient-Specific Auxetic Stent Formation

The final stage of the stent design process is to model the auxetic stent that conforms to the reconstructed arterial geometry. This step utilizes the 3D artery model generated by the volumetric segmentation and reconstruction pipeline.

Using Autodesk Inventor, mesh-to-face feature-scripts are used to wrap the auxetic lattice structure around the 3D arterial model. This process involves mapping each auxetic unit's geometric parameters (strut width t and the strut angle variable θ) to the local curvature and dimensions of the artery's surface.

Finite Element Analysis Optimization

Once the patient-specific auxetic stent design was completed, it was evaluated using in-silico techniques, specifically finite element analysis (FEA) to test its performance and further optimize the model. The 3D CAD model from the auxetic unit wrapping step was imported into the simulation environment. The artery was treated as a hyperelastic material to mimic vascular tissue and the stent was assigned nitinol's superelastic properties [21,22].

Then, the stent was virtually deployed inside the artery model. The radial expansion against the vessel wall was simulated along with the cyclic loading from pulsatile blood flow. The stent geometry was meshed with fine tetrahedral elements to capture localized stress at auxetic unit junctions. A frictional contact model was used to represent the mechanical interaction between the stent and artery wall.

A transient pulsatile Computational Fluid Dynamic (CFD) simulation was run over several cardiac cycles to mimic in-vivo conditions [23,24]. This allowed for the observation of the stent's expansion and ability to maintain the vessel's open diameter without over-compressing healthy arterial regions. Final refinements to the model allow the stent to conform closely to the patient-specific arterial geometry

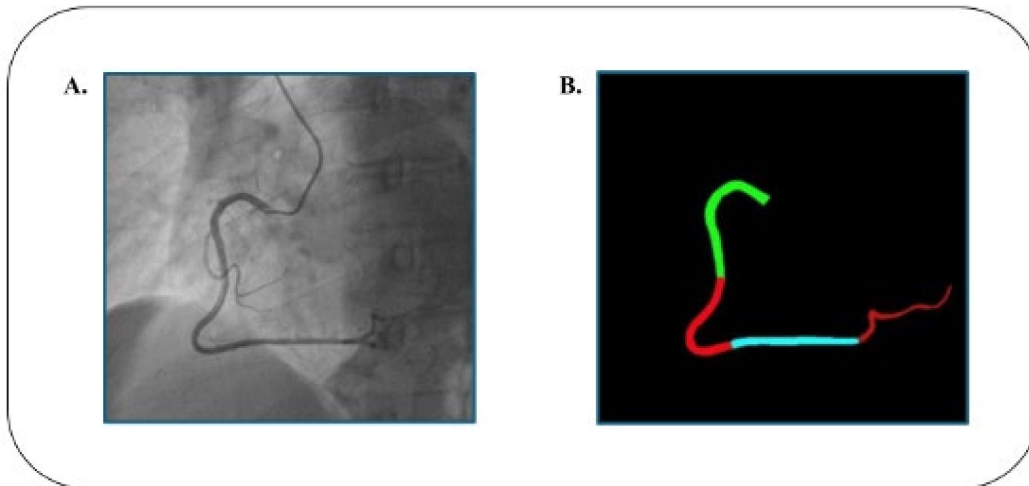


Figure 2. Illustrates the input (left) and output (right) images from the model. Green indicates a healthy region, red indicates a stenotic region, and blue indicates a thrombotic region.

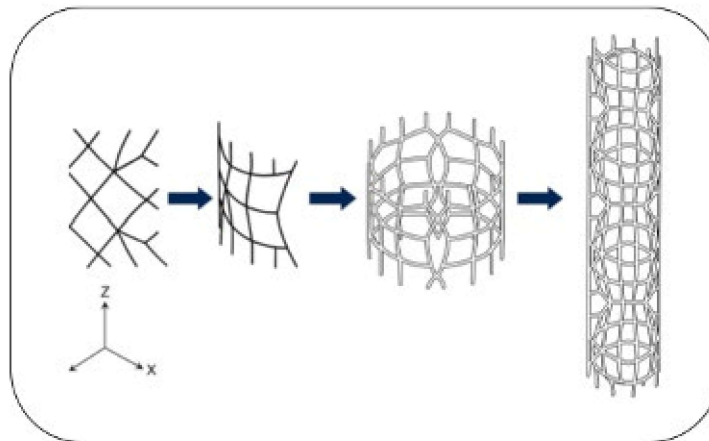


Figure 3. Illustration of auxetic wrapping process. Pictured on the left is a singular auxetic cell. Pictured on the right is a complete stent that is ready for parameter adjustment.

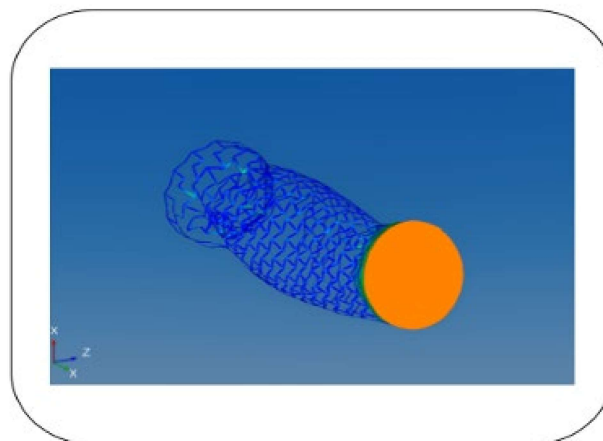


Figure 4. FEA optimized auxetic stent model. Note how the auxetic stent has varying diameters and shape formations unlike conventional stents.

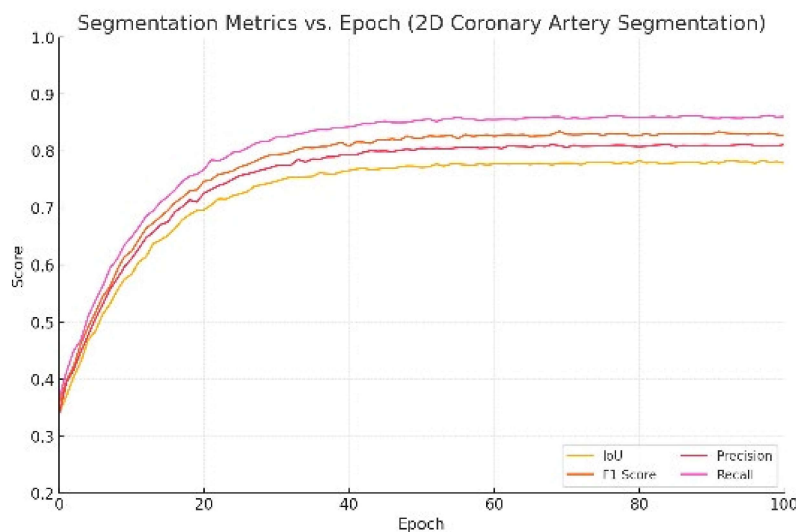


Figure 5. (Segmentation Metrics vs. Epoch): IoU, F1 score, precision, and recall all increase sharply in early training and stabilize at high values, reflecting balanced segmentation performance.

while maintaining structural integrity, thereby reducing the risk of restenosis or vessel damage.

Mesh & Stress Machine Learning Model

A third machine learning model was developed to predict stress distribution in the stent's CAD mesh and to automatically reinforce regions that were considered "weak." This process reduced the need for repeated manual finite element simulations during design optimization.

To prepare the training data, multiple stent designs were created with varying auxetic unit t and θ parameters. Each design was analyzed in ANSYS, where a stress test was performed to record nodal coordinates and the resulting von Mises stress values. These datasets linked the stent's geometry to its mechanical performance under simulated physiological conditions.

A Graph Neural Network (GNN) [25,26,27] was trained to recognize patterns between the mesh geometry and the stress distribution across the stent. Input features included node positions while the output was the predicted von Mises stress for each node.

Once trained, the model could take a FEA optimized patient-specific stent mesh, predict stress concentrations, and identify areas at risk of mechanical failure. Using this information, it automatically added material to those regions in the CAD model to strengthen the stent without significantly reducing auxetic properties.

RESULTS

Coronary Artery Segmentation Model Metrics

The 2D coronary artery segmentation model performance was evaluated using standard semantic segmentation metrics. The model had a final validation accuracy of 93%, IoU of 0.78, F1 score of 0.83, precision of 0.81, and recall of 0.86.

Auxetic Unit Strength Testing

Six distinct unit-cell geometries were modeled to evaluate the mechanical performance of each against deflection stresses. The chosen designs were: re-entrant, tri-chiral, tetra-chiral, hexa-chiral, tri-anti-chiral, and tetra-anti-chiral.

FEA simulations were conducted using the in-silico testing simulation shown in Figure 8 (A). The setup consisted of a three-point bending configuration, with two fixed supports (P2 and P3) and a single central load application point (P1) exerting displacement compression along the stent's longitudinal axis.

Von Mises stress distributions were taken from each simulation to assess structural performance under bending loads. Shown in Figure 8 (B), stress values were normalized to the maximum recorded across all simulations. Color gradients along the stent indicate areas of peak strain accumulation.

Among all tested configurations, the re-entrant geometry exhibited the lowest peak von Mises stress of 178 mPA (Figure 9), suggesting superior load distribution and deformation

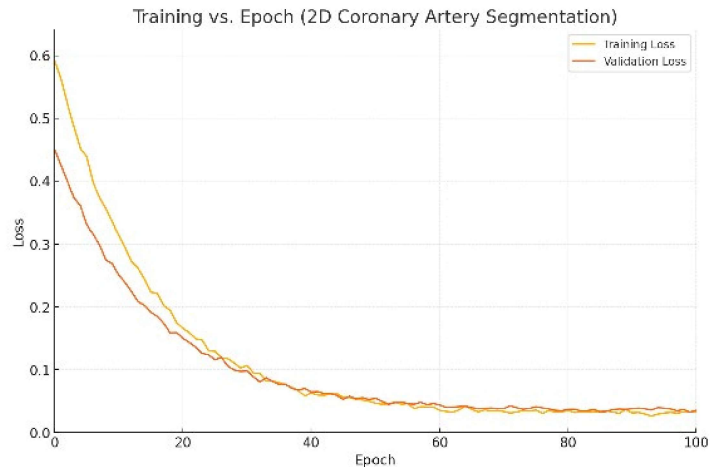


Figure 6. (Training vs. Validation Loss): Both training and validation loss decrease steadily over 100 epochs, converging to low final values (~0.03), indicating stable learning and minimal overfitting.

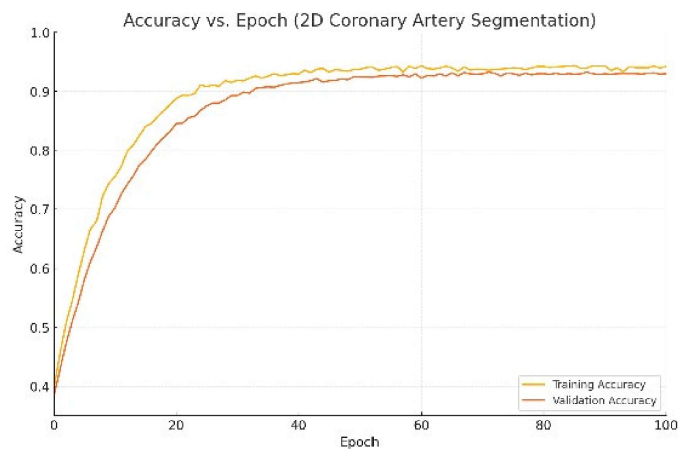


Figure 7. (Training vs. Validation Accuracy): Accuracy improves rapidly during early epochs before plateauing near 93%, with training and validation curves closely aligned throughout training.

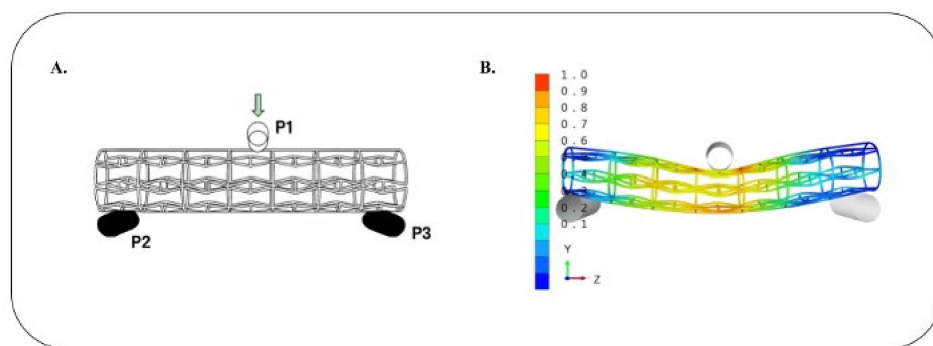


Figure 8. Re-entrant geometry is shown in both images. Von Mises stress testing simulation shown on the left. Stress gradient of the stent shown on the right.

tolerance compared to the other auxetic designs.

Finite Element Analysis Validation

A comparative FEA was conducted to validate the performance of the auxetic stent design against a conventional stent. The analysis measured radial principal stress, peak stress, and fatigue life of both stent designs.

As shown in Figure 10, the auxetic stent had a significantly lower radial principal stress across all measured vessel positions. The peak principal stress of the conventional stent exceeded 220 MPa while the auxetic stent maintained a maximum of approximately 150 MPa. This meant that the auxetic stent exerted less stress onto the arterial wall, reducing the risk of vascular wall damage.

Peak stress and fatigue life metrics (Figure 11A, 11B) confirm the performance advantage further. The ML-optimized auxetic stent had a 32% reduction in peak stress compared to the conventional design. This reduction directly contributed to an extended fatigue life as well, with the auxetic stent achieving an estimated five times longer cycle life of 50 million cycles versus 10 million cycles for the conventional stent.

Blood Flow Testing

Further in-silico testing between auxetic and conventional stents was conducted measuring blood flow, using ANSYS Fluent for computational fluid dynamics (CFD) simulations. Blood flow was evaluated under physiological pulsatile

pressure conditions across the stented arterial segment. As shown in Figure 12, the conventional stent showed greater fluctuations in flow rate and a lower average blood flow, mainly due to uneven radial expansion creating local turbulence and zones of disturbed flow. In contrast, the auxetic stent demonstrated a higher and more stable average flow rate throughout the cardiac cycle, indicating reduced turbulence and lower overall flow resistance.

DISCUSSION

The primary objective of this study was to determine whether a machine learning optimized bi-stable auxetic stent could address the limitations of conventional vascular stents, particularly in reducing restenosis and improving long-term vessel patency. The results from FEA and CFD simulations demonstrate that the auxetic design significantly reduces peak principal stresses, enhances fatigue life, and improves blood flow stability compared to traditional stents.

A key finding was the lower and more uniform radial principal stress profile in the auxetic stent, which aligns with evidence that minimizing localized high-stress concentrations can reduce vessel wall trauma—a known factor in neointimal hyperplasia and restenosis [9,8]. Additionally, the five-fold increase in predicted fatigue life suggests that the proposed design may withstand physiological loading for longer durations without failure, consistent with prior work showing the clinical relevance of fatigue-resistant stent geometries [21].

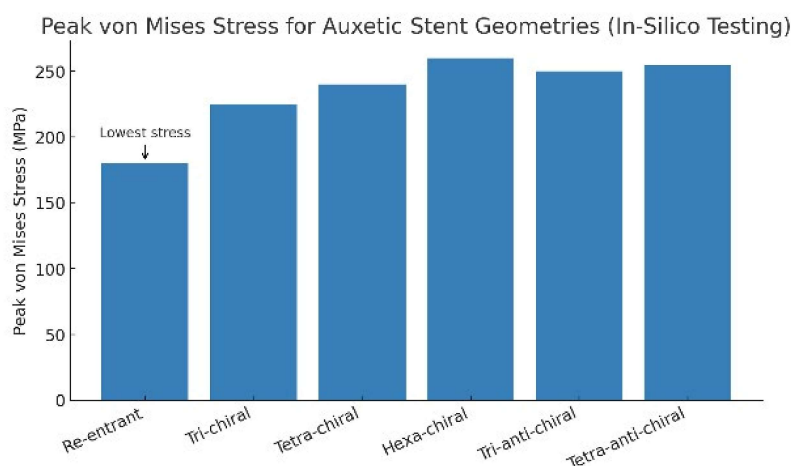


Figure 9. Bar Graph of the six various auxetic designs plotted against their respective deflection stress force.

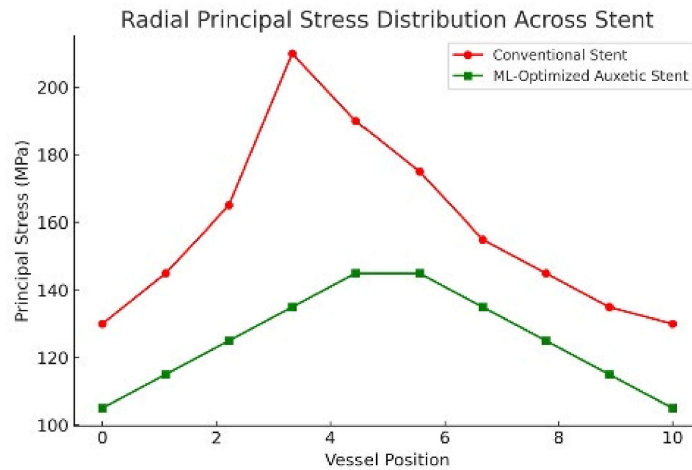


Figure 10. Graph showing Radial Principal Stress Distribution along Vessel Position of both conventional stents and auxetic stents.

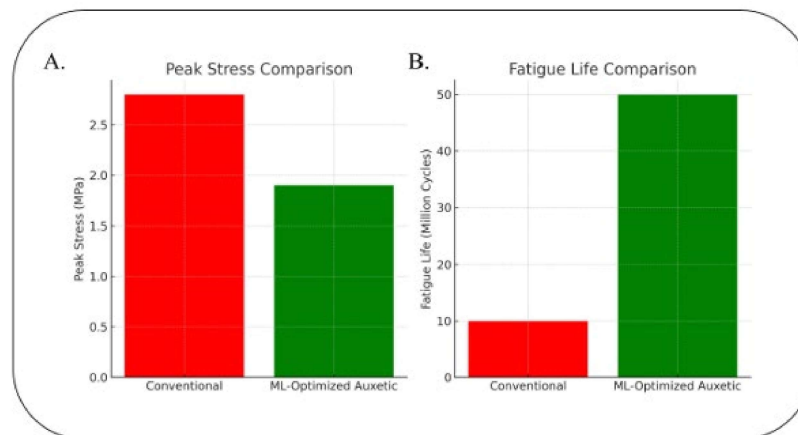


Figure 11. Pictured on the left is a Peak Stress graph comparing conventional v.s. Auxetic stents. On the right, is a Fatigue Life Comparison graph.

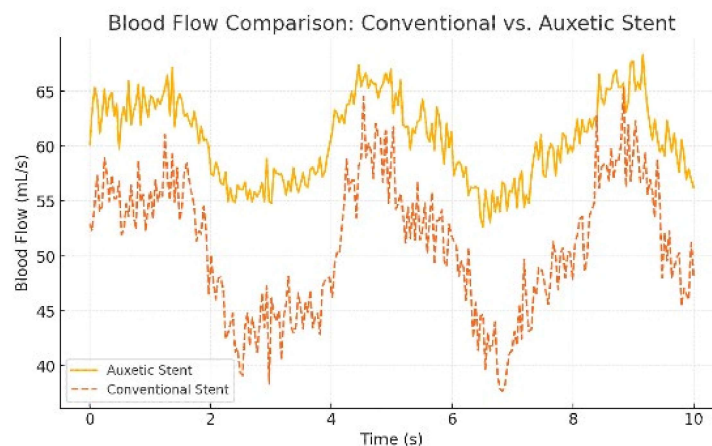


Figure 12. Blood Flow Comparison generated from computational fluid dynamic techniques.

The CFD simulations revealed that the auxetic stent maintained higher and more stable blood flow rates. This is significant, as favorable hemodynamic conditions are associated with reduced platelet aggregation and inflammatory responses [23,24]. The more consistent flow pattern observed here supports the hypothesis that geometry-tailored radial expansion can help maintain physiological shear stress, thereby mitigating restenosis risk [7].

In the context of the broader literature, previous studies on auxetic structures have highlighted their potential in biomedical implants due to their unique deformation mechanics and ability to distribute stresses evenly [10,12]. However, most prior research has been limited to theoretical modeling or non-biological applications. This study advances the field by integrating auxetic mechanics with ML-driven patient-specific customization, an approach that parallels recent trends in precision medicine and computational design optimization [15].

Limitations of this work include its in-silico nature; biological processes such as inflammatory response and calcification were not directly modeled. Furthermore, while the ML algorithm effectively optimized the auxetic lattice based on imaging-derived vessel geometries, its performance is dependent on the quality and diversity of training data. Future research should include in-vitro bench testing with pulsatile flow systems and in-vivo preclinical studies to validate the mechanical and biological performance of the design under physiological conditions.

CONCLUSION

These findings show that an ML-optimized bi-stable auxetic stent design has the potential to surpass conventional stent limitations through improved conformability of complex-shaped vascular geometries, lower localized mechanical stress, higher fatigue life, and higher blood flow rate. Patient-specific imaging and computational optimization aligns with the trend of personalized medicine in interventional cardiology procedures.

Both the hemodynamic and mechanical advantages have a promising future potential to lower rates of restenosis and enhance late lumen durability in patients. Some of the areas requiring future development include optimization of the fabrication process to allow for clinical

scalability, experimental model validation of the design, and incorporation of the optimization pipeline into current interventional planning pipelines. Translated effectively into clinical practice, the new design has the potential to offer more durable and physiologically compatible solutions to complex coronary disease patients.

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